

Thanya Mysore Santhosh

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EDUCATION

Northeastern University
Master of Science in Data Science

Sep. 2024 – Apr. 2026
Boston, MA

Vidyavardhaka College of Engineering
B. Tech, Computer Science & Engineering

Aug. 2019 – Jul. 2023
Mysuru, India

PUBLICATION

NU_DeepHealthNLP at #SMM4H-HeaRD 2026

[Read Publication](#)

Entity-Conditioned Generation and a Four-Stage Pipeline for Automated SOAP Note Generation. ACL 2026 Workshop. 1st place, SMM4H-HeaRD 2026.

TECHNICAL SKILLS

Languages & ML Frameworks: Python, SQL, JavaScript, TypeScript, PyTorch, Hugging Face, Scikit-learn, XGBoost

Generative AI & Agentic Systems: Mistral, BioMistral, LLaMA, Gemini, OpenRouter, Retell AI, QLoRA, vLLM, Voice AI, RAG, GraphRAG, LangGraph, MCP, A2A, FAISS, Neo4j, multi-agent orchestration

Applications, Data & Cloud: React, Express, FastAPI, Flask, Streamlit, PostgreSQL, MongoDB, BigQuery, Snowflake, GCP, Vertex AI

MLOps, Testing & Analysis: Docker, Kubernetes, GitHub Actions, CI/CD, pytest, SHAP, BERTopic, GLiNER

EXPERIENCE

AI Engineer, Saturn Tech

Jul. 2026 – Present

- Contribute to a **multi-tenant AI recruitment platform** (React, TypeScript, Express, PostgreSQL) supporting candidate, job, pipeline, and AI-enabled recruitment workflows
- Develop and validate **LLM-powered candidate screening** and **voice AI interview workflows** across automated recruitment stages
- Test and debug end-to-end AI workflows across frontend, backend APIs, integrations, and asynchronous processing to validate **reliable application behavior**
- Collaborate directly with the CEO and recruiters to gather requirements, understand end-to-end recruitment workflows, and translate business needs into features and technical requirements

Research Assistant, Clinical NLP, Northeastern University

Jan. 2026 – Apr. 2026

- Fine-tuned **Mistral-7B** with **QLoRA** and built a **four-stage clinical generation pipeline** across 8,529 training dialogues, incorporating entity conditioning, retrieval, prompting, and verification
- Ranked **1st place** at **SMM4H-HeaRD 2026** and published findings in the **ACL 2026 workshop proceedings**

Data Scientist, 66degrees

May 2023 – Jul. 2024

- Developed and tuned customer-churn models (XGBoost, Random Forest, Logistic Regression), improving ROC-AUC from 0.78 to 0.85 through behavioral feature engineering and hyperparameter optimization
- Built K-Means segmentation workflows to identify high-value, inactive, and at-risk customer cohorts, applying SHAP to explain individual predictions for business stakeholders
- Partnered directly with client stakeholders and cross-functional delivery teams in weekly working sessions to gather requirements, present findings, and translate retention challenges into data-science solutions

PROJECTS

Enterprise Knowledge Orchestrator

[GitHub](#)

- Architected a **multi-agent RAG system** using **Model Context Protocol**, orchestrating specialized research, code, and writer agents across 4 independent tool servers
- Built a **GraphRAG pipeline combining vector search** with **multi-hop knowledge graph traversal**, plus an AI governance layer with audit logging and PII detection/redaction

Screen Agent: Vision-Grounded UI Automation

[GitHub](#)

- Diagnosed and quantified a real limitation in vision-language model UI grounding — measured a VLM's coordinate predictions collapsing to viewport-center regardless of true element position (**35–57% localization error**)
- Designed and implemented a **DOM-based grounding backend** as a fix, reading real element positions from the browser instead of estimating them visually, achieving **100% task success** across **5 verified runs**

BreastCare: Clinical Decision Support

[GitHub](#)

- Developed **CoxPH survival models** on 2,509 METABRIC patients for overall and relapse-free survival, achieving held-out **C-indices** of **0.625** and **0.609** within a three-model clinical decision-support platform
- Contributed to a three-model clinical ML platform spanning diagnosis, risk stratification, and survival analysis, with project-level results of **98.8% diagnosis accuracy**, **1.000 ROC-AUC**, and **90.9% three-class risk accuracy**